

RESEARCH ARTICLE

Road Traffic Crash Severity Analysis: A Bayesian-Optimized Dynamic Ensemble Selection Guided by Instance Hardness and Region of Competence Strategy

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ABSTRACT Millions of lives are lost in road accidents annually, underscoring the severity of traffic incidents. Despite numerous studies on crash severity forecasting with static ensemble machine learning, the potential of Dynamic Ensemble Selection (DES) for combining classifiers and improving performance remains relatively unexplored in this field. In contrast to previous research focused on static ensemble machine learning models, our study introduces an interpretable Bayesian Optimized Dynamic Ensemble Selection strategy, reinforced with instance hardness and region of competence considerations, for crash risk assessment. Utilizing six algorithms – Random Forest, Extra Tree, Cat Boost, Light GBM, KT Boost, and XGBoost – optimized through cross-validation, we employed bagging as a hybrid ensemble selection criterion to ensure ensemble diversity for DES models. Addressing class imbalance with cost-sensitive learning, we evaluated six DES algorithms, ‘KNORA-E, KNORA-U, DES-KNN, DES-RRC, DES-KL, and META-DES’ using a pool of input classifiers, based on key metrics such as Balanced Accuracy Score, Sensitivity, G-mean, and Matthews Correlation Coefficient. Our results revealed that the KNORA-E ensemble with Cat Boost achieved the highest performance, with a Balanced Accuracy (BCA) of 69.3%, Matthews Correlation Coefficient (MCC) of 0.36, and a G-mean score of 69.10%. DES-RRC demonstrated notable performance across all base classifiers, particularly in addressing imbalanced classes, and DES-RRC with LightGBM emerged as the second-best performer. SHAP analysis for the KNORAE-Cat Boost model identified key factors in crash severity as, road user gender, airbag deployment, vehicle age and road surface conditions. This approach could be a valuable tool for informed decision-making and targeted interventions.

INDEX TERMS Bayesian optimization, cost-sensitive learning, crash severity analysis, dynamic ensemble selection, DES-RRC-light GBM, KNORAE-Cat Boost, SHapley Additive exPlanations.

I. INTRODUCTION

Due to the growth in motorization and travel demand, traffic accidents are worsening. Predictions suggest that by the next decade or so, they might outpace cancer as the sixth leading global cause of death. Despite improvements in mass trans-

sit, the involvement of artificial intelligence, and intelligent transportation systems, road safety remains a significant challenge, with millions of fatalities and over 50 million injuries annually worldwide [1]. In the last couple of years, there is a modest decrease in the annual number of road traffic deaths to 1.19 million, indicating that measures to increase road safety are having an effect. But in order to meet the target of halving road traffic fatalities and injuries by 2030, immediate action

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is required because the cost of mobility is still much too high [2]. In the light of this, researching the causes of traffic accidents and estimating their severity have gained popularity in the modern period. Many smart applications have been created and modeled by researchers to address concerns of safety, performance measures, and convenience. When state-of-the-art technology is unable to prevent traffic accidents and their consequences, safety concerns continue to be a major obstacle [3].

In recent years, machine learning (ML) has been widely applied for accurate crash severity forecasting by capturing complex relationships without relying on preset assumptions. Using a variety of statistical techniques, researchers [4], [5], [6], [7], [8], [9], [10] have spent years attempting to establish a connection between the severity of crash injuries and risk factors such as highway design, vehicle characteristics, timing, crash type, traffic regulations, driver traits, weather, and time-related factors. They have identified several important influence factors for predicting traffic accidents, as well as several methods for determining their severity. Road safety analysis relies heavily on an understanding of accidents and crucial components when analyzing traffic crash severity prediction. Authorities can effectively lower crash rates by using the useful information provided by improved predictive models. Researchers have tried to establish a connection between variables and accident severity by employing a variety of statistical methodologies. The emphasis has recently switched to ensemble learning methods in an effort to improve predictive performance when assessing crash severity and related risks [11], [12], [13], [14], [15]. Essentially, ensemble machine learning has proven to be superior over statistical approaches both empirically and conceptually. In order to produce a better forecast estimate and range, ensemble learning integrates the results of several machine learning and statistical techniques.

Investigating literature, tree-based and boosting-based static ensemble methods have been used extensively in crash risk assessments over the past few years due to their reliable performances, quick speeds, scalability, along with optimization and interpretation techniques to get around the inherent limitations of conventional statistical techniques. The majority of these earlier studies [4], [5], [8], [13], [16], [17], [18], [19], [20] used these machine learning approaches to forecast accident severity based on injury level or fatality rate. In order to accurately forecast and examine the influencing elements in traffic accidents, Elyassami et al. [16] used tree-based techniques such as decision trees, random forests, and gradient boosted trees to explore the causes and severity of crashes using real-world data and proposed that for predicting the most important factors causing accidents, tree-based gradient models demonstrated better accuracy. Iranitalab and Khat-tak [21] investigated four statistical and machine learning techniques namely multinomial logistic regression, KNN, SVM, and RF to forecast the severity of traffic crashes and predicts in their findings that multinomial logistic regression

displays the lowest estimations, and k-nearest neighbor performed the best in both major and minor crashes. Sangare et al. [22] introduced a groundbreaking framework that combines the descriptive power of the Gaussian Mixture Model and Support Vector Classifier. Their innovative approach supplies the SVC with mean vectors from the GMM model, leading to superior performance over traditional statistical methods in their experimental results.

Guo [23] in his study on Colorado crash data using the XG Boost model revealed that the driver attributes, older pedestrian characteristics, and vehicle maneuvers are the pivotal factors influencing severity levels. Bokaba et al. [24] in their Comparative Study of Machine Learning Classifiers for Modeling Road Traffic Accidents used Naive Bayes, logistic regression, k-nearest neighbor, AdaBoost, support vector machines, and random forests to compare machine learning algorithms for forecasting traffic accidents. They found out that the RF model, when applied by linked equations with multiple imputations, produced the best results in empirical analyses for crash severity predictions. Zhang et al. [25] employed a combination of RF, XGB, NB, KNN, and BLR classifiers alongside the Boruta Algorithm (BA), an inherent feature selection approach, to identify significant attributes in their hybrid feature selection method for assessing the severity of vehicular accident injuries. The analysis highlighted vehicle type, month of the year, the age of the driver, and road alignment as the most significant variables.

Nonetheless, there has been a rise in interest in using static ensemble learning techniques for crash severity analysis in transportation safety in recent years. These techniques have become well-known because they are straightforward and efficient, producing encouraging outcomes and insights into enhancing road safety measures. These traditional approaches, however, frequently depend on a single machine learning algorithm, which may reduce forecast accuracy and ignore intricate data connections.

In contrast, using the capabilities of several algorithms in static ensemble learning, particularly by using techniques like stacking, has been shown in some experiments to significantly increase prediction performance. For example, Niyogisubizo et al. [26] presented a novel stacking framework on a hybrid of gradient and adaptive boosting approaches with multilayer perceptron, improving the accuracy of determining the severity of collision injuries. Comparably, Ahmad, et al. [27] highlighted the effectiveness of stacking in their study, emphasizing its superior prediction accuracy, particularly in identifying critical factors like average offset distance to fixed objects and commercial driveway density. These studies highlight the practical significance of static ensemble learning, in particular stacking, in improving prediction accuracy and suggesting practical road safety measures.

In machine learning for crash severity analysis, besides conventional statistical models, ensemble learning approaches like bagging, boosting, and heterogeneous methods such

as stacking or blending are widely employed to enhance predictive accuracy by combining multiple base learners. Static ensemble approaches, however, may perform less well when handling complex and dynamic data distributions.

DES methods address this problem by adaptively selecting the most capable learners for each prediction task. DES techniques have emerged as more precise and trustworthy intelligent techniques in credit fraud detection, biomedical and environment related fields [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39].

Specifically, ensemble learning can be categorized as static or dynamic based on how classifiers are selected. The process as shown in Fig. 1 involves three stages: generation, selection, and integration. Generation establishes a pool of base classifiers, selection picks one or an ensemble, and integration fuses their predictions for the final outcome. Selection, a critical phase, is focused on either static (SS) or dynamic selection (DS), where static selection relies on average base classifier performance and dynamic selects a classifier based on competence [40], [41], [42].

Due to their efficacy and efficiency, DES classifiers have garnered a lot of attention. Because 'DES' dynamically discovers a classifier for each test data separately, it effectively helps to simultaneously harness the gains from less biased and low-variance predictions supplied by diverse ML approaches [40], [41]. Moreover, DES approaches are more efficient and adaptable since they dynamically identify an ensemble of classifiers for each test data, as opposed to searching for a single classifier for the entire dataset, aiming to improve prediction accuracy based on their aptitude for particular problem instances [43], [44], [45], [46].

In the domain of crash severity research, the intrinsic complexity of incident data poses a significant challenge, attributed to a wide range of contributing factors, including weather, road conditions, vehicle features, and driver conduct. Moreover, the problem of class imbalance, in which the number of minor or non-severe accidents is greater than the number of severe ones, emphasizes how important it is to have DES techniques in order to successfully manage this complex heterogeneity.

While Khattak et al. [14] employed advanced dynamic ensemble selection approach in a classification scenario to forecast injury severity; however, the utilization of DES strategies, coupled with an in-depth exploration of why these techniques enhance classification performance. For example, taking into account, concepts such as 'region of competence', which pertains to the feature space where a given classifier excels, and 'instance hardness', which denotes the difficulty in classifying specific instances in the dataset – this remains either unexplored or minimally investigated in predicting crash frequency on roadways, to the best of our knowledge.

In light of the aforementioned literature's glaring gaps, the primary insights of this research are highlighted as follows:

- Guided by the DES framework, we propose a rigorous and robust DES strategy enhanced with instance

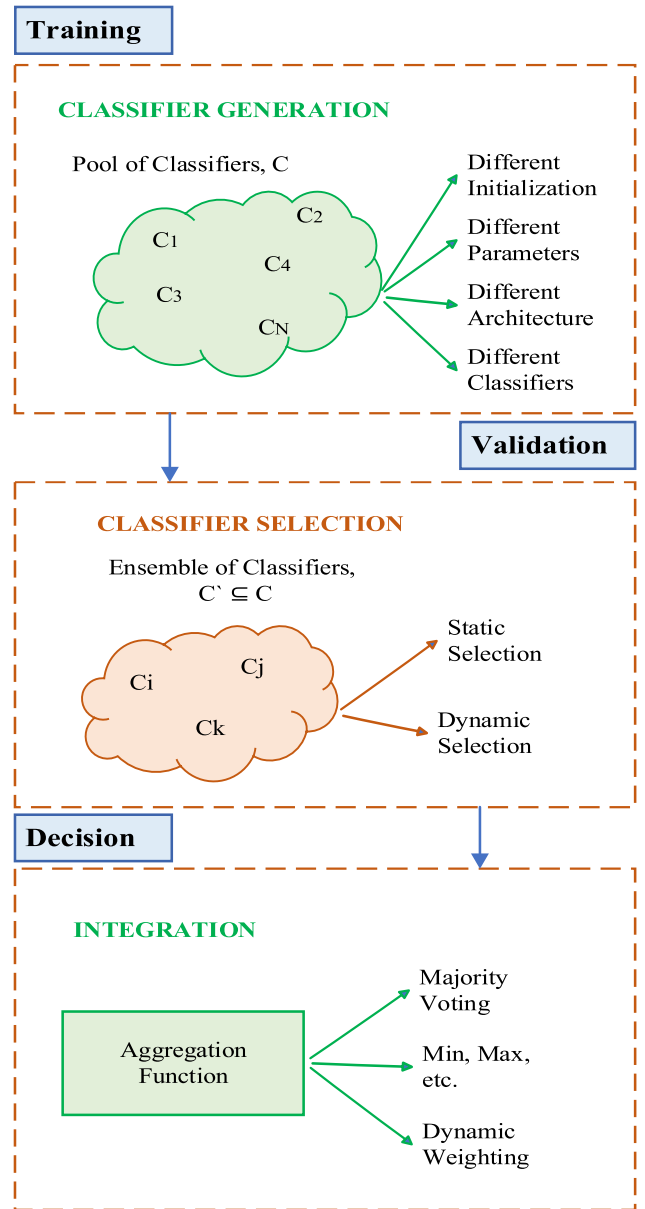


FIGURE 1. Possible phases of ensemble selection process.

hardness and region of competence, aimed at improving the predictive performance for crash severity analysis. We investigated six state of art DES methods including KNORA-E, KNORA-U, DES-KNN, DES-RRC, DES-KL, and META-DES to develop an accurate classification prediction model.

- Specifically, to facilitate the selection of competent and diverse classifiers, the diversity within the DES algorithms is achieved through a two-step process: optimizing the pool of static ensemble classifiers namely – Random Forest, Extra Tree, Cat Boost, LightGBM, XGBoost, and KT Boost – and utilizing bagging to construct a diverse ensemble of classifiers with hybrid ensemble selection criteria.

- Moreover, cost-sensitive learning mechanism, utilizing class weights is used to address the imbalanced class distribution during optimization of the candidate pool classifiers with hyper-opt Bayesian optimization strategy, while applying the expected improvement acquisition function to enhance the performance of base classifiers and proposed DES models.
- In addition, a competence driven weighted majority vote scheme is introduced at the decision integration stage for predicting the performances, enhancing the reliability of the ensemble predictions.
- Finally, the framework is extended by incorporating explainability using SHAP (SHapley Additive exPlanations) analysis, which facilitates the interpretation of influential features and enhances the transparency of the developed models.

To fulfill the outlined research objectives, we utilize a detailed dataset spanning from 2009 to 2015 sourced from the National Automotive Sampling System (NASS) General Estimates Data System (GES) of the United States. Additionally, we conduct a comparative analysis of the ensemble pool of classifiers with and without employing DES algorithms. The evaluation of model performance is conducted based on Balanced Accuracy, Sensitivity, G-mean Score, and MCC. Our findings demonstrate that the integration of more accurate, reliable, and robust intelligent DES techniques yields superior performance prediction capabilities compared to individual statistical or ensemble Machine Learning methods. The structure of paper is organized as follows: Section II presents the materials and methods employed, Section III outlines the experimental results and subsequent discussions, and Section IV explores the limits of the study and provides a summary of the main conclusions.

II. METHODOLOGY

This paper employs a comparative methodology to analyze the performance of six state-of-the-art DES algorithms against baseline static ensemble strategies, aiming to enhance predictive performance of Crash Risks in terms of Balanced Accuracy, Sensitivity, G-Mean score and Matthew's correlation coefficient (MCC). The analysis was conducted using open-source Python packages, including Scikit-learn, DESlib, SHAP, etc. These tools were essential for implementing the proposed machine learning models, carrying out data preprocessing, and performing the analysis. Fig. 2 illustrates the overall processing pipeline of the methodology, which proceeds as follows:

- Data pre-processing, involving dataset integration, manual feature subset selection from NASS-GES data, removing inconsistencies such as duplicates and unknown counts, and checking and removing strongly correlated features (if any).
- Splitting the data into training-validation and test sets.

- Fine-tuning and validating model hyperparameters using a stratified 5-fold cross-validation strategy on the training-validation set via Bayesian optimization.
- Utilizing a Bagging classifier to construct diverse ensembles of classifiers for each Base classifier (RF, Extra Tree, Cat-Boost, Light GBM, XG Boost, and KT Boost) fitted with optimal parameters and cost-sensitive approach with class-weights to handle target variable imbalance.
- Experimenting with six DES algorithms (KNORA-E, KNORA-U, DES-KNN, DES-RRC, DES-KL and META-DES) enhanced with Instance Hardness and Region of Competence using Bayesian optimization.
- Comparing the performance of the aforementioned algorithms and further interpreting the DES algorithm showing best predictive performance with SHAP for major feature incorporation into risk assessment.

A. DATA DESCRIPTION AND PRE-PROCESSING

The dataset that is being used pertains from the NASS-GES, which is derived from police reports relating to car accidents between 2009 and 2015. Because it integrates information from trauma centers and engineering, the attributes encompass diverse facets such as patient vital signs, hospital records, laboratory tests, treatment orders, and vehicle manufacturing and model details. Based purely on transportation engineering knowledge, a subset of independent variables representing main factors contributing to accidents was identified. The attributes of dataset, classified into categories as personal, road-related, environmental, temporal and vehicular characteristics, summarized in Table. 1.

The dataset, which includes 9,369 documented collisions, provides an extensive information repository that is essential for comprehending accident dynamics. To ensure data integrity, unknown data entries were excluded from the dataset, resulting in a final set of 9,167 accident records.

To enhance predictive modeling suitability, categorical features underwent label encoding, ensuring compatibility with machine learning algorithms. A dependent variable, 'Severity Level,' was defined to describe the extent of damage, ranging from property damage only (PDO) to injuries. Due to low counts of severe and fatal injuries, these were combined with minor/possible injuries, framing the analysis as a binary classification problem with classes for property damage and injuries.

Furthermore, major aspects contributing to road traffic safety were modified for everyday usage. To ensure clarity and prevent misinterpretation of attributes, we have adjusted the names ('INJ-SEV' has been renamed to 'Severity_Level', 'LGHT-CON' to 'Light_Condition', 'V-ALCH-I' to 'Alcohol_Involvement') of key factors contributing to road traffic safety, making them more accessible for everyday understanding.

Moreover, we used Pearson correlation in our research, carried out in a python programming environment, to look

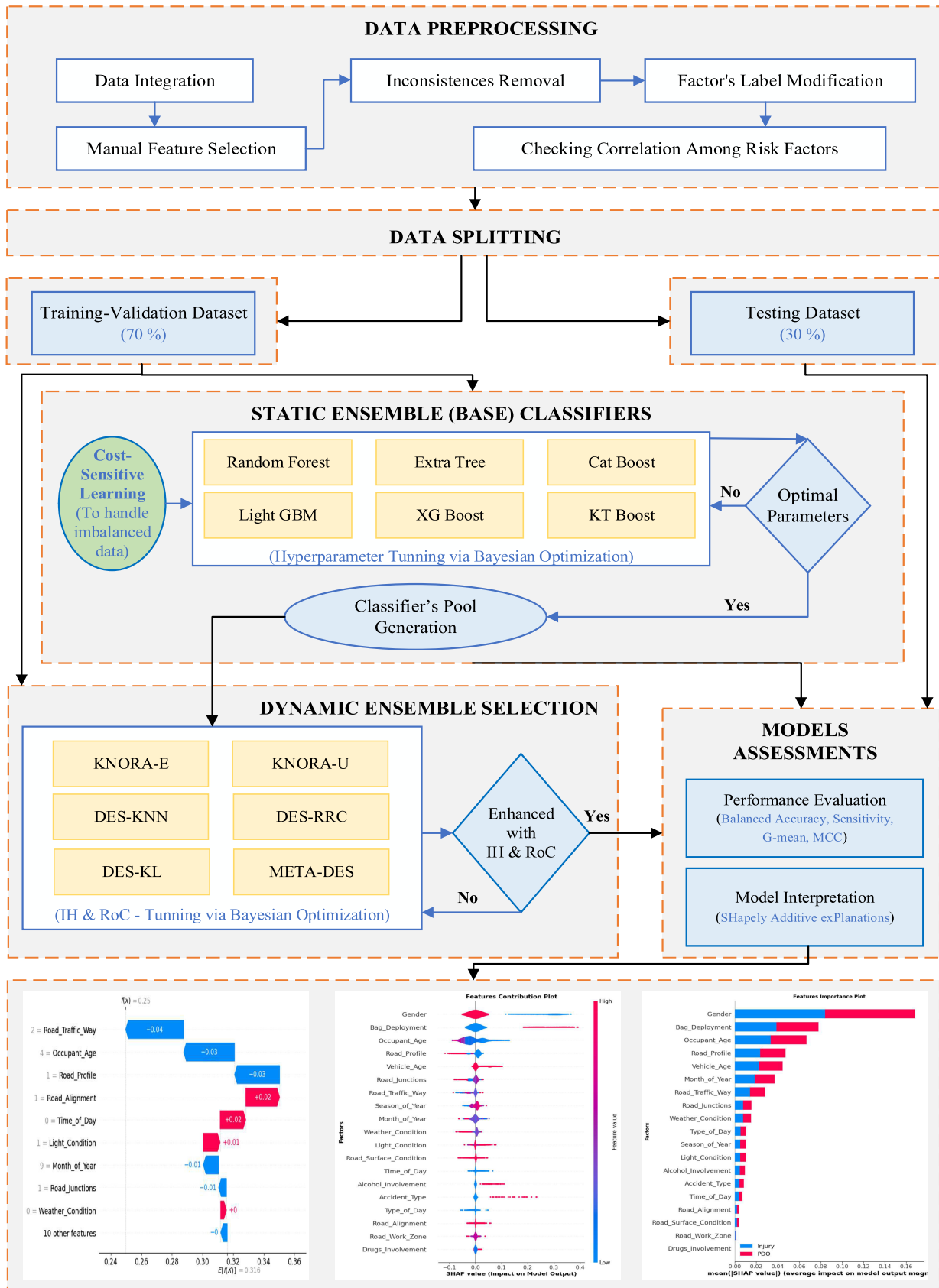


FIGURE 2. Structural view of proposed framework for analyzing severity of crashes.

into possible relationships between different parameters that were captured in crash data. The results, illustrated in Fig. 3, revealed predominantly weak positive correlations, ranging from 0.0004 to 0.35, among the risk factors examined [47].

Even though the time of day and light conditions at night, showed a substantial negative correlation coefficient of -0.68 , we intentionally kept these characteristics for additional modeling due to their moderate relationship, understanding contextual dependencies and acknowledging their potential impact on model outcomes if integrated. Therefore, depending on particular circumstances and contextual variations, all these factors continue to be crucial considerations for our modeling efforts.

Data splitting has been carried out for analysis by segregating the dataset of 9167 counts into mainly two sets: the 'Training and Validation' subset of 70% and the 'Testing' dataset of 30 %. To preserve an equal proportion of the classes for the training-validation and testing sets for the experiments, 6416 numbers in the training-validation set and 2751 numbers in the testing dataset maintained, respectively.

B. STATIC ENSEMBLE CLASSIFIERS

Static Ensemble Selection involves choosing a consistent subset of classifiers from a pool of available options to merge predictions. This method maintains a fixed selection of classifiers throughout the learning process. Once the subset is established, they collectively form an ensemble model, and their predictions are aggregated to reach final decisions. Typically, this approach relies on heuristics, such as opting for classifiers with strong individual performance or diverse characteristics. However, its limitation lies in its lack of adaptability to fluctuations in dataset dynamics or classifier performance, as the chosen subset remains unchanged.

In our investigation, although we employed a range of classifiers, including Decision Tree, AdaBoost, Random Forest, Extra Tree, Multi-Layer Perceptron (MLP), Cat Boost, SVM, Light GBM, XGBoost and KT Boost within a homogeneous static ensemble framework for classification tasks. However, upon evaluating their performances, we found certain algorithms namely Random Forest, Extra Tree, Cat Boost, Light GBM, XGBoost and KT Boost, exhibited notably better results. Consequently, we have chosen these algorithms for further consideration as base classifiers within DES approaches.

1) RANDOM FOREST

Random Forest is actually a bagging technique characterized by bootstrapping and random sampling criteria, developed by Breiman [48]. This technique involves the creation of number of decision trees trained on subsets of training dataset, wherein each tree contributes to the final prediction through an aggregation process, typically averaging the outputs or utilizing voting mechanisms. By leveraging this ensemble of decision trees, Random Forest aims to enhance predictive

accuracy while mitigating overfitting, thereby reducing overall variance in the model.

2) EXTRA TREE

Extra Tree, also known as Extremely Randomized Trees, is an ensemble learning technique that constructs multiple decision trees with random splits to reduce variance and improve generalization performance. Unlike traditional decision trees, Extra Tree employs intentional randomization of the splitting criteria, by selecting 'the number of attributes for each node and the minimum sample size for splitting the node. By incorporating this intentional randomness, Extra Tree aims to make the model less sensitive to noise and irrelevant features. Instead of bootstrap duplicates, Extra Tree utilizes the entire original learning sample to reduce bias. The final prediction is made through majority voting among the aggregated classifier trees [49].

3) CATEGORICAL BOOSTING

The gradient-boosting decision tree method with advancement, introduced by Friedman [50], called as Cat Boost, emerges as a refined iteration capable of effectively managing highly nonlinear data due to its robust learning capabilities. By harnessing the Gradient Boosting Decision Tree (GBDT) technique, Cat Boost amalgamates numerous decision trees to construct a model of high predictive accuracy [51]. Notably, the parameters are updated in the gradient decent direction of the loss function which enhances model performance. At the heart of gradient boosting lies the objective of creating additional root learners that optimally align with the negative gradient of the loss functions [17]. With its specialized focus on handling categorical features, Cat Boost contribute to heightened model performance and enhanced training efficiency.

4) LIGHT GRADIENT BOOSTING MACHINE

Light GBM is a new representative of the boosting ensemble developed by Microsoft in collaboration with Peking University's research group [52], leverages the GBDT algorithm effectively. It adopts a leaf-wise growth strategy with depth constraints, along with Gradient-based one-side Sample preparation (GOSS) and Exclusive Feature Bundling (EFB) techniques [51]. Designed for efficiency and scalability, LightGBM ensemble system aggregates multiple decision trees, with each tree framework trained to anticipate the residuals of earlier models, resulting in faster training speed and reduced memory usage compared to traditional gradient boosting methods, to generate a well-generalized final prediction.

5) EXTREME GRADIENT BOOSTING

The XG Boost algorithm, introduced by Chen and Guestrin [53] in 2016, represents an ensemble technique that combines multiple learning algorithms to enhance predictive accuracy. This approach creates a series of decision trees to

TABLE 1. Description of selected input features used for binary classification.

Variable Name	Risk Factor	Variable Description	Value Counts
Severity Characteristics	Severity_Level	0; Property Damage Only 1; Injury {Possible: 1391, Minor: 812, Serious: 239, Fatal: 31 & Severity Unknown: 75}	6619 2548
	Occupant_Age (Years)	0; ≤ 20 , 1; 20-29, 2; 30-39, 3; 40-49, 4; 50-59, 5; ≥ 60	918, 2172, 1830, 1811, 1446, 990
Personal Characteristics	Gender	0; Female, 1; Male	2612, 6555
	Alcohol_Involvement	0; Alcohol Not Involved, 1; Alcohol Involved	8958, 209
	Drugs_Involvement	0; Drugs Not Involved, 1; Drugs Involved	9143, 24
	Bag_Deployment	0; Bag Deployed, 1; Bag Not Deployed	8088, 1079
Temporal Characteristics	Season_of_Year	0; Winter, 1; Spring, 2; Summer, 3; Fall	2895, 2237, 2353, 1682
	Month_of_Year	0; January, 1; February, 2; March, 3; April, 4; May, 5; June, 6; July, 7; August, 8; September, 9; October, 10; November, 11; December	816, 613, 771, 672, 794 830, 755, 768, 773, 909 757, 709
	Type_of_Day	0; Weekend, 1; Weekdays	1107, 8060
	Time_of_Day	0; Night, 1; Day	2041, 7126
	Weather_Condition	0; Clear, 1; Rain, 2; Snow, Hail, 3; Fog, Smog, Cloudy, 4; Others (Blowing sand, Severe cross wind)	6870, 863, 255, 1107 72
Environmental Characteristics	Light_Condition	0; Day light, 1; Dark with road lights, 2; Dark without road light, 3; Others (dawn, dusk, unknown light)	7054, 1247, 595 271
	Road_Profile	0; Level, 1; Grade, 2; Others (Up/Down Hill, Sag, Hillcrest)	6309, 1369, 1489
Road Characteristics	Road_Alignment	0; Straight, 1; Curve, 2; Others (non-traffic way, unknown, driveway access)	8462, 461, 244
	Road_Surface_Condition	0; Dry, 1; Wet, 2; Others (Sand, Dirt, Slush, unknown)	7503, 1616, 48
	Road_Traffic_Way	0; 2-way Not-divided, 1; 2-way Divided, 2; Others (1-way, ramp, non-traffic way)	2249, 5278 1640
	Road_Junctions	0; No-Intersection, 1; Intersection, 2 – Others (unknown, not-reported)	5145, 3830 192
	Road_Work_Zone	0; No-work Zone, 1; Work Zone, 2 – Others (not-reported)	8721, 317 129
	Vehicle_Age (Years)	0; ≤ 9 , 1; 10-19, 2; 20-29, 3; ≥ 30	5893, 2950, 288, 36
	Accident_Type	0; Not Rollover, 1; Rollover, 3; Others (unknown)	9086, 72, 9

iteratively refine the model, with each new tree aiming to match the residuals, calculated as the gradient of the loss function, from the preceding iteration. Consequently, each decision tree learns from the errors of its predecessors, facilitating the gradual improvement of the model. Furthermore, the loss function of XGBoost incorporates a regularization term and a training loss to promote model diversity and mitigate overfitting.

6) COMBINED KERNEL AND TREE BOOSTING

Combining Kernel and Tree Boosting (KT Boost) techniques entails capitalizing on the respective strengths of both methodologies to enhance predictive performance. Kernel methods, characterized by their ability to capture complex nonlinear relationships using kernel functions, complement the boosting algorithms' proficiency in capturing interactions and feature importance through ensembling multiple decision trees. In the KT Boost algorithm, the initial training phase involves learning both a tree and a Reproducing Kernel Hilbert Space (RKHS) ridge regression function. This model is subsequently optimized using either the functional Newton's method or gradient de-scent. The selection of the

ensemble base learner is based on minimizing empirical risks while maximizing predictive accuracy. During each iteration, a base learner is chosen from two functional classes: firstly, the continuous functions in the RKHS with higher regularity determined by the kernel function, and secondly, the discontinuous functions represented by trees. This strategic combination of diverse base learners enhances the model's ability to learn functions with varying degrees of regularity more effectively [54].

7) CLASSIFIER POOL GENERATION

Classifier pool generation is pivotal step for dynamic ensemble methods, where static ensemble members are selected based on current data or their existing performance. It involves strategies for selecting, or updating ensemble members to enhance performance over time. Hence, classifier pool generation is essential for adapting to changing conditions and maintaining predictive accuracy. Pool of classifiers for our study actually falls in three steps, (1) Tuning with optimal parameters for static ensemble methods, (2) Preprocessing for class imbalance, and finally (3) Creating pool of classifiers for diversity in DES methods.

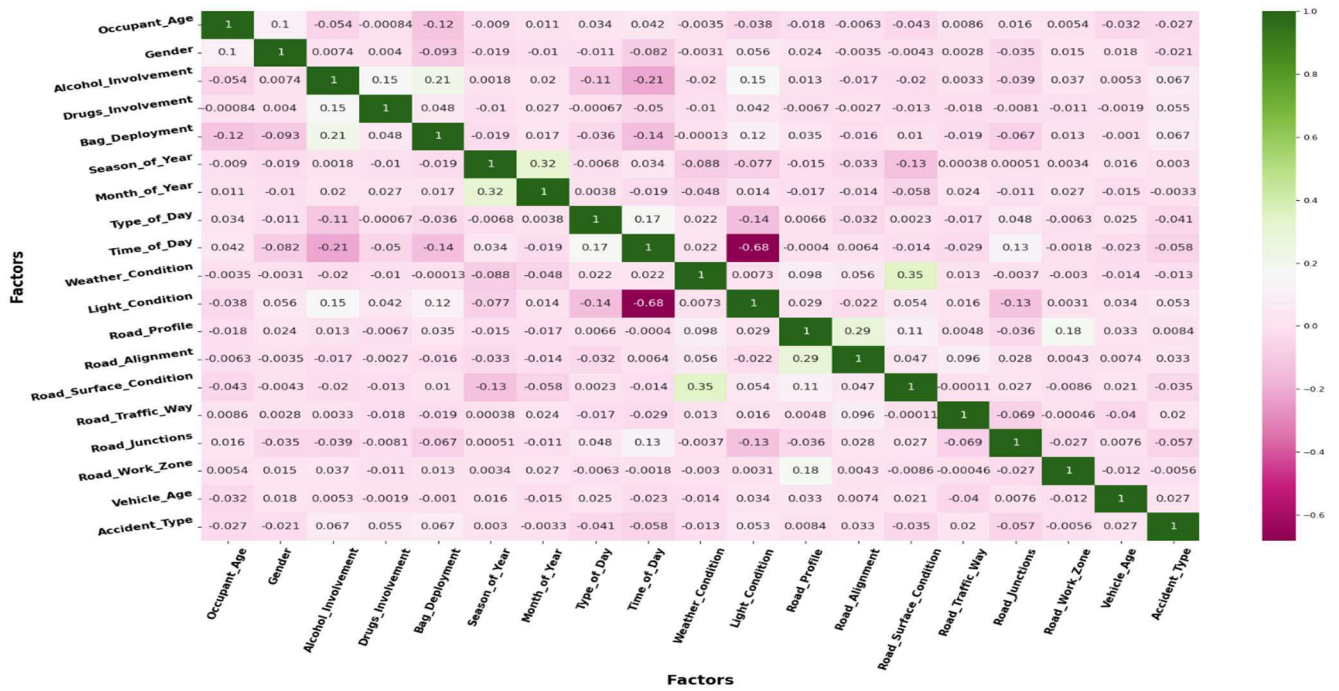


FIGURE 3. Pearson correlation matrix highlighting relationships among risk factors.

- 1) Hyperparameter tuning of base classifiers; Prior to creating the pool of classifiers, each base classifier (static ensemble method) undergoes optimization using Bayesian optimization with the Hyper-opt open-source library. Bayesian optimization is preferred due to its ability to simultaneously assess different sets of hyperparameters, unlike grid search and random search, which explore combinations sequentially. By focusing on promising areas of the search space, Bayesian optimization improves model performance. Unlike grid search, which may not be ideal for evaluating complex goal functions or adaptive exploration, Bayesian optimization offers a more thoughtful and flexible approach. This makes it particularly valuable in scenarios with tight tuning budgets and significant function evaluations.
- 2) Preprocessing for Imbalance Data; the cost-sensitive mechanism utilizing class weights, is employed to address the imbalanced nature of data when fitting base classifiers to create a pool of classifiers. By doing so, the base classifiers mitigate learning challenges arising from imbalanced data. Additionally, the randomized nature of preprocessing enhances the diversity of the classifier pool. During pool generation, this preprocessing yield improved results compared to static ensemble methods, influencing the performance of dynamic ensemble techniques [55], [56]. Class weights as cost-sensitive learning, are chosen due to their simplicity and ease of implementation. Unlike oversampling and under sampling, which modify the

dataset, class weights directly adjust the learning process to account for class imbalance.

While oversampling and under sampling provide additional training data, class weights avoid introducing extra samples. Although cost-sensitive learning may not be as effective as oversampling or under sampling for severe class imbalance, it remains a valuable alternative, especially when dataset modification is impractical. Furthermore, cost-sensitive learning with class weights helps prevent overfitting by adjusting the loss function during training, making them a strong tool for handling class imbalance in machine learning models.

- 3) Creating pool of base classifiers; After optimizing with optimal parameters and utilizing cost-sensitive learning to handle class imbalance, a pool of classifiers is generated for each base static ensemble method using the Bagging technique. Bagging constructs a diverse ensemble by randomly selecting subsets of training data, enhances robustness, reduces overfitting, and improves overall predictive performance ensuring each classifier trains on a distinct subset [29].

C. DYNAMIC ENSEMBLE SELECTION ALGORITHMS

Once the base classifier's pool is created, the next step is to choose classifiers ensemble. DES within Multiple Classifier Systems (MCS) stands out as a promising approach for enhancing pattern recognition accuracy. Unlike traditional ensemble models with fixed classifier compositions, DES dynamically selects base classifiers based on each new

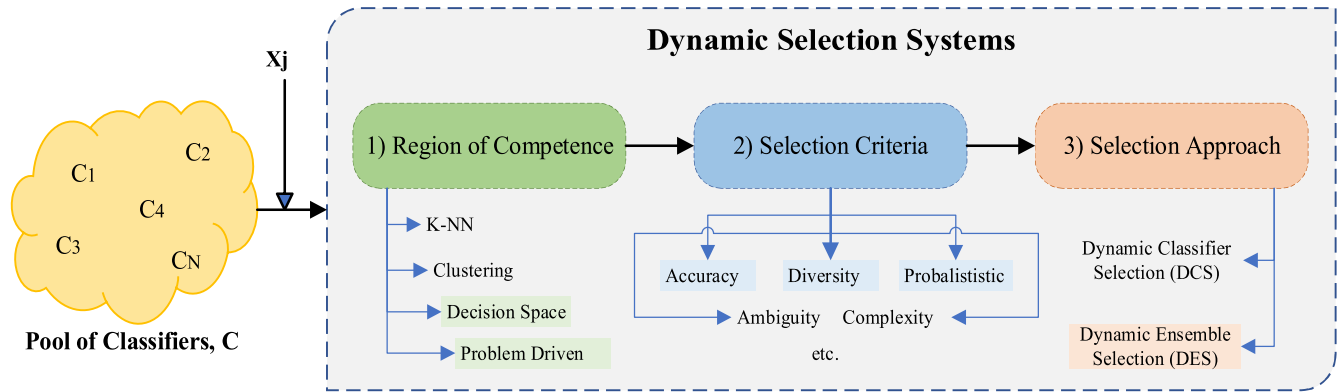


FIGURE 4. Taxonomy of three-step dynamic selection systems.

sample. This adaptability hinges on a three-step process as shown in Fig. 4, defining a region of competence by evaluating similarity between training data and new samples, applying selection criteria to assess classifier performance, and employing a selection approach to choose classifiers for prediction. By adapting to sample characteristics, DES enables ensemble models to improve prediction performance significantly compared to traditional static ensemble approaches [41].

The region of competence assumes paramount importance, delineating the specific segment of the feature space where each classifier exhibits superior predictive capabilities. The reliability of classification systems based on DES hinges on the precise evaluation of competence. However, inaccurate assessments may lead to the selection of suboptimal classifiers within the ensemble for a given region, potentially impairing system performance [43], [57], [58].

Additionally, in-stances situated within regions of indecision are frequently misclassified by other pattern recognition methods due to their tendency to exhibit a high level of instance difficulty [59]. Concurrently, instance hardness serves as a vital measure, capturing the intricacy involved in classifying individual samples. Through the intricate integration of these factors into the optimization framework, the selection method strives to achieve a delicate balance in accuracy scores across a diverse ensemble of classifiers.

During the DES phase, the objective is to choose proficient and diverse classifiers for each test sample. To accomplish this, an optimization-driven approach utilizing Bayesian optimization is employed considering two crucial parameters, namely the region of competence and instance hardness, for every DES method examined in the study. Subsequently, the performance of each classifier is enhanced using the Expected Improvement acquisition function to improve balanced accuracy. If a classifier enhanced with RoC and IH, deemed competent, it is then incorporated into the ensemble.

Despite employing a multitude of DES methodologies in our analysis, such as DES-P, KNOP, DES-Exp, DES-Log, KNORA-E, KNORA-U, DES-KNN, DES-RRC, and META-

DES. We have opted for six cutting-edge DES algorithms, namely KNORA-E, KNORA-U, DES-KNN, DES-RRC, DES-KL, and META-DES in our research publications due to their significant performances.

1) KNORA-E

The KNORA-Eliminate (KNORA-E) technique in Dynamic Ensemble Selection identifies classifiers that achieve perfect predictions within a defined region of competence, termed as local Oracles. It selects all classifiers that exhibit 100% accuracy within this region and aggregates their decisions using majority voting. If no classifier meets this criterion, the size of the competence region decreases iteratively until competent classifiers are found. KNORA-E thus emphasizes the selection of highly reliable classifiers for ensemble composition, ensuring robust predictions in dynamic environments [40], [41].

2) KNORA-U

The KNORA-Union method within DES selects all classifiers that correctly identify at least one sample within the region of competence. It allows base classifiers to contribute multiple times to the voting scheme if they make correct predictions for multiple instances within the region. The number of votes assigned to a particular base classifier is equivalent to the number of correctly predicted samples within the region of competence. For example, if a base classifier correctly predicts the label for three samples within the region, it gains three votes for the majority voting scheme. The votes accumulated by all base classifiers are aggregated to derive the ensemble decision. KNORA-U thus prioritizes the inclusion of classifiers demonstrating competence within specific regions, enhancing the ensemble's predictive accuracy in dynamic scenarios [40], [41].

3) DES-KNN

DES-KNN, Dynamic Ensemble Selection K-Nearest Neighbors, aims to enhance ensemble classifier performance by considering both accuracy and diversity of base classifiers.

It utilizes the k-NN algorithm to define the competence region, selecting the N most accurate classifiers within it. Through a Markov Decision Process formulation, Q-learning is applied to learn an effective selection policy. DES-KNN is particularly effective on imbalanced datasets and integrates well with data preprocessing methods. In a nutshell DES-KNN prioritizes accuracy and diversity, while dynamically selecting the best classifiers for ensemble improvement [41], [60].

4) DES-RRC

The DES-RRC, Dynamic Ensemble Selection with Randomized Reference Classifier [46] is a technique designed to improve ensemble classifier systems by carefully selecting base classifiers for ensemble inclusion. It leverages the concept of a randomized reference classifier (RRC) to evaluate the performance of individual base classifiers in comparison to a randomized baseline. DES-RRC enhances ensemble classifiers by selectively choosing base classifiers based on their performance relative to RRC. It evaluates each base classifier's competence using a Gaussian potential function ($K(x_i, x_j)$) and a source of competence (Csrc). This assessment helps mitigate the influence of individual data points in the dataset, estimating the base classifiers' ability to classify query instances. Those exceeding the competence level of the RRC, which is set as $1/L$ (where L is the number of classes), are included in the ensemble. This strategy prioritizes proficient classifiers, enhancing ensemble classifier effectiveness [41].

5) DES-KL

The DES-KL, Dynamic Ensemble Selection with Kullback Leibler Divergence method assesses the competence of base classifiers within an ensemble by leveraging Kullback Leibler (KL) divergence, measuring the difference between class supports provided by base classifiers and those from a random classifier (RC). DES-KL estimates the competence of classifiers from an information theory perspective. It dynamically selects fine-tuning layers based on this divergence, enhancing model adaptation and accuracy in transfer learning tasks. By computing KL divergence for each instance in the data, DES-KL identifies base classifiers with positive divergence values, indicating competence in making accurate predictions. This process streamlines ensemble composition by focusing on classifiers demonstrating proficiency in specific instances; thereby, optimizing model performance [41].

6) META-DES

META-DES, Meta-Learning Dynamic Ensemble Selection is a machine learning technique which works by dynamically selecting the most competent classifiers and optimizing ensemble performance through meta-learning. It enhances ensemble classifier performance by evaluating the competency of individual base classifiers. It utilizes meta-learning, where various criteria are represented as meta-features. These

features inform a meta-classifier, trained to assess classifier competence. During the generalization phase, meta-features extracted from query instances guide the meta-classifier in determining the suitability of base classifiers for ensemble inclusion [61].

D. MODEL ASSESSMENTS

1) MODEL'S PERFORMANCE EVALUATION

Given the inherent class imbalance within our dataset, it becomes imperative to consider the importance of various performance metrics in effectively evaluating predictions. Specifically, when dealing with imbalanced data, the significance of metrics such as Balanced Accuracy score, Sensitivity, G-mean score, and MCC are heightened due to their ability to provide a more nuanced assessment of model performance. These metrics are systematically assessed for both static and dynamic ensemble methods, with computations based on the confusion matrix, shown in Fig. 5.

To ensure comprehensive understanding of these performance metrics, the terminology employed is defined as: True Positive (TP) denotes the total count of predictions accurately identifying a crash instance as resulting in injury. True Negative (TN) represents the total count of predictions accurately identifying crash instances as resulting in property damage only. False Positive (FP) indicates the total count of predictions incorrectly classifying property damage instances as injury. Conversely, False Negative (FN) denotes the total count of predictions inaccurately classifying injury instances as property damage. Below are concise descriptions of each evaluation metric used to assess injury severity prediction.

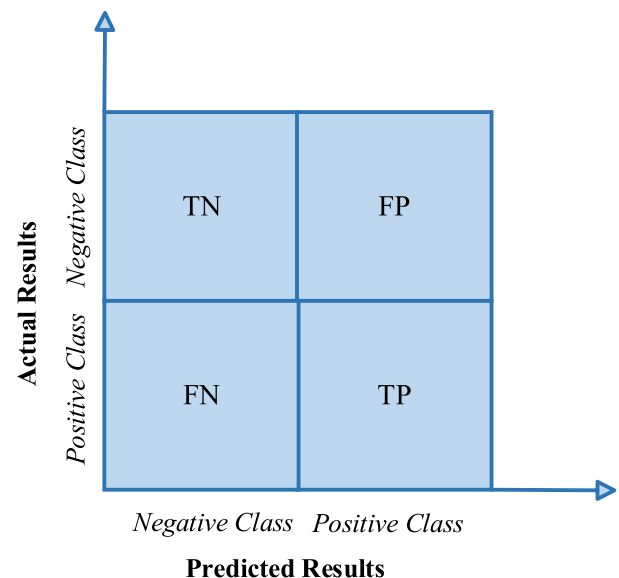


FIGURE 5. Confusion matrix for binary classification problem.

Balanced Accuracy score, as in (1) takes into account the imbalance between the classes by calculating the average of sensitivity and specificity, offering a reliable measure of overall model accuracy that is not overly influenced by class

distribution. It is an improved version of the conventional accuracy metric designed to work better with unbalanced datasets.

$$Bal.Acc. = \frac{1}{2} \left(\frac{T_P}{T_P + F_N} + \frac{T_N}{T_N + F_P} \right) \quad (1)$$

Sensitivity also known as recall or true positive rate, in the context of imbalanced datasets, is crucial as it measures the model's capability to detect the positive instances (minority class) without misclassifying them as the negative instances (majority class), as demonstrated in (2).

$$Sensitivity = \left(\frac{T_P}{T_P + F_N} \right) \quad (2)$$

G-mean Score as shown in (3), are also important in the context of imbalanced data as they measure the geometric mean of sensitivity and specificity, providing a balanced assessment of the model's performance across both classes.

$$G - mean = \sqrt{\left[\frac{T_P}{T_P + F_N} \right] \times \left[\frac{T_N}{T_N + F_P} \right]} \quad (3)$$

Matthew's correlation coefficient (MCC) is highly valuable for imbalanced datasets as it considers all four quadrants of the confusion matrix (TP, TN, FP, FN), providing a more comprehensive evaluation of predictive performance that is robust to class distribution, as depicted in (4).

$$MCC = \frac{T_P T_N - F_P F_N}{\sqrt{(T_P + F_P) \cdot (T_P + F_N) \cdot (T_N + F_P) \cdot (T_N + F_N)}} \quad (4)$$

The MCC quantifies the correlation between predicted and actual classifications, yielding a value between -1 and $+1$. A coefficient of $+1$ signifies an ideal prediction, while 0 represents an average performance, and -1 indicates a poor prediction.

Incorporating these performance metrics ensures a more thorough and accurate assessment of model effectiveness in handling imbalanced data; thereby, enabling informed decision-making and enhancing the reliability of predictions regarding crash injury severity.

2) MODEL'S INTERPRETATION

Utilizing the SHapley Additive exPlanations (SHAP) [62] method enhances the interpretability of machine learning models employed in crash injury severity prediction. SHAP quantifies the significance of each feature, capturing interactions and dependencies within the model. This comprehensive approach provides fair insights into the contribution of individual features to predictions, both locally and globally. Locally, SHAP offers precise explanations of how each feature impacts specific predictions, while globally, it highlights overarching patterns in feature importance across the dataset.

Rooted in game theory principles, SHAP's Shapley values accurately depict each feature's marginal contribution to model output, accommodating intricate interactions. With its

additive attribution strategy, SHAP ensures thorough consideration of each feature's contribution, facilitating accurate and reliable interpretations of model predictions. Overall, SHAP emerges as an indispensable tool for researchers and practitioners seeking to elucidate the inner workings of machine learning models in crash injury severity prediction.

III. EXPERIMENTS AND DISCUSSIONS

This study employed crash severity forecasting utilizing data from the NASS-GES between 2009 and 2015. All injuries encompassing 'minor, major, serious, unknown, and fatal injuries' were consolidated into a single dependent variable denoted as 'Injury-Severity' the minority class, while property damage only constituted the majority class. Fatal injuries were included within the injury count to facilitate comprehensive model interpretation and machine learning predictions. To ensure data quality, the initial dataset underwent preprocessing to address unknown or unidentified counts. Subsequently, the dataset was divided into training/validation (70%) and test sets (30%), for testing/performance analysis, while ensuring consistent analysis and prediction accuracy. The primary purpose of partitioning the dataset into training-validation and testing sets is to mitigate the risk of overfitting.

A. BASE CLASSIFIERS 'STATIC ENSEMBLE SELECTION'

The optimal hyperparameters for the base 'Static Ensemble' models are determined using Bayesian optimization with the hyper-opt strategy. This method systematically explores predefined hyperparameter combinations based on cross validation performance, making it a novel technique for tuning hyperparameters in machine learning tasks within the crash severity domain. Bayesian optimization guides the search by selecting hyperparameter combinations that exhibit promising performance during cross-validation. In our experiments, we prioritize hyperparameter combinations that maximize the overall balanced accuracy, calculated as the average balanced accuracy across all folds in a Stratified 5-Fold Cross Validation setup. Table. 2 showcases the range and optimal value for each classifier's primary hyperparameters.

Hyperparameter tuning via Bayesian optimization is applied to all the base classifiers to identify the optimal hyperparameters. Balanced accuracy, which accounts for sensitivity and specificity of severity labels, is selected as the Expected Improvement acquisition function, particularly pertinent for the imbalanced dataset. The test set remains unseen and is reserved exclusively for evaluating the performance of trained models.

Following hyperparameter optimization, fitting of every static ensemble method is done with the pure training dataset by utilizing class weight mechanism to counteract the imbalanced nature of data; thereby, alleviating learning difficulties stemming from data imbalance. Subsequently, a pool of classifiers is generated for each base static ensemble method employing the Bagging technique; then, these

TABLE 2. Hyperparameter optimization of Base ‘static ensemble’ classifiers.

Parameters	Hyper-opt Space	Optimal Values
Random Forest Hyperparameter Optimization		
max_depth	(1, 21)	17
min_samples_split	(2,11)	5
min_samples_leaf	(1,11)	1
n-estimators	(10, 500)	266
Extra Tree Hyperparameter Optimization		
max_depth	(1, 21)	15
min_samples_split	(2,11)	3
min_samples_leaf	(1,11)	8
n-estimators	(10, 500)	101
Cat Boost Hyperparameter Optimization		
iterations	(10, 200)	131
learning-rate	(-1, 1)	0.052173008
depth	(2,20)	14
Light GBM Hyperparameter Optimization		
num_leaves	(10, 150)	122
max-depth	(2,20)	11
learning-rate	(-1,1)	0.113349417
XG Boost Hyperparameter Optimization		
learning-rate	(-1,1)	0.123949611
max-depth	(2,20)	4
n_estimators	(10,200)	54
KT Boost Hyperparameter Optimization		
max_depth	(2,20)	3
min_sample_leaf	(1,5)	4
n_estimators	(10,200)	162
theta	(1,10)	6

hyperparameter-optimized machine learning pools of classifiers are employed across the six DES algorithms.

B. DYNAMIC ENSEMBLE SELECTION

Region of competence and instance hardness are crucial in DES, significantly impacting the effectiveness of classification systems utilizing DES methodologies. Region of competence identifies feature space areas where each classifier excels in prediction. Accurate evaluation of this region is vital for selecting the most suitable classifiers within the ensemble for a given data segment. Similarly, instance hardness indicates the complexity of classifying individual samples, with instances in regions of indecision typically demonstrating heightened difficulty.

Although some studies [31], [63] have utilized the region of competence ‘k’ determining through experimentation with varying values such as 3, 5, 7 up to 19 for all DES algorithms during cross-validation, our approach innovatively utilizes Bayesian optimization. In our methodology, we optimize DES considering instance hardness and region of competence, employing balanced accuracy as the expected acquisition function to enhance performance.

The range of values considered for Instance Hardness (IH_rate) and RoC (k) in the Bayesian optimization process with Hyper-opt space, are [0.1 to 0.5] & [2 to 25] respectively. The DES optimization framework intricately integrates these factors, aiming for a delicate balance in accuracy scores across diverse classifier ensembles. Effective consideration of instance hardness and region of competence enhances DES systems’ decision-making abilities, ultimately improving overall classification performance. Fig. 6 displays the 3D scatter plot illustrating the hyperparameter tuning values obtained through Bayesian optimization for the primary DES methods with maximum performances alongside the corresponding base classifier.

C. MODEL COMPARISON AND PERFORMANCE EVALUATION

In this investigation, the classification of positive and negative instances was designated as ‘Injury’ and ‘Property Damage Only’, respectively. The performance of various models was assessed using several metrics including balanced accuracy, sensitivity, G-mean Score, and MCC. These metrics were evaluated for each static ensemble model as well as DES algorithms utilizing a pool of base learners.

1) PERFORMANCE OF STATIC ENSEMBLE CLASSIFIERS

While employing a stratified 5-fold cross-validation methodology on the training-validation dataset, devoid of consideration for class weights, multiple static ensemble classifiers were trained. Among them, those models demonstrating superior accuracy in classifying Crash injury severity were identified and selected as base classifiers for subsequent DES models. The results for stand-alone static ensemble methods are represented in the Table. 3.

It has been evaluated from Table. 3 that, among the static ensemble classification models, the Categorical Boosting classifier exhibited superior performance with a balanced classification accuracy of 63.12%, sensitivity of 74%, MCC of 0.291, and a G-mean score of 58.5%. Following closely, the Light Gradient Boosting Machine model achieved a balanced classification accuracy of 63.09%, sensitivity of 74%, MCC of 0.289, and a G-mean score of 58.5%. While KT Boost classifier demonstrated a higher sensitivity score of 75% and an MCC value of 0.30, it lagged behind in terms of balanced accuracy score and G-mean score, which provide more comprehensive evaluations of the model’s performance across imbalanced classes. Nonetheless, all chosen static ensemble models exhibited the ability to classify injury severity with an accuracy exceeding 60%.

2) PERFORMANCE OF DYNAMIC ENSEMBLE CLASSIFIERS

Subsequently, the Random Forest, extremely randomized Tree, Categorical Boosting, Light GBM, XG Boost, and KT Boost pools were employed in tandem with KNORA-E, KNORA-U, DESKNN, DES-RRC, DES-KL, and META-DES algorithms. Table. 4 illustrates the performance

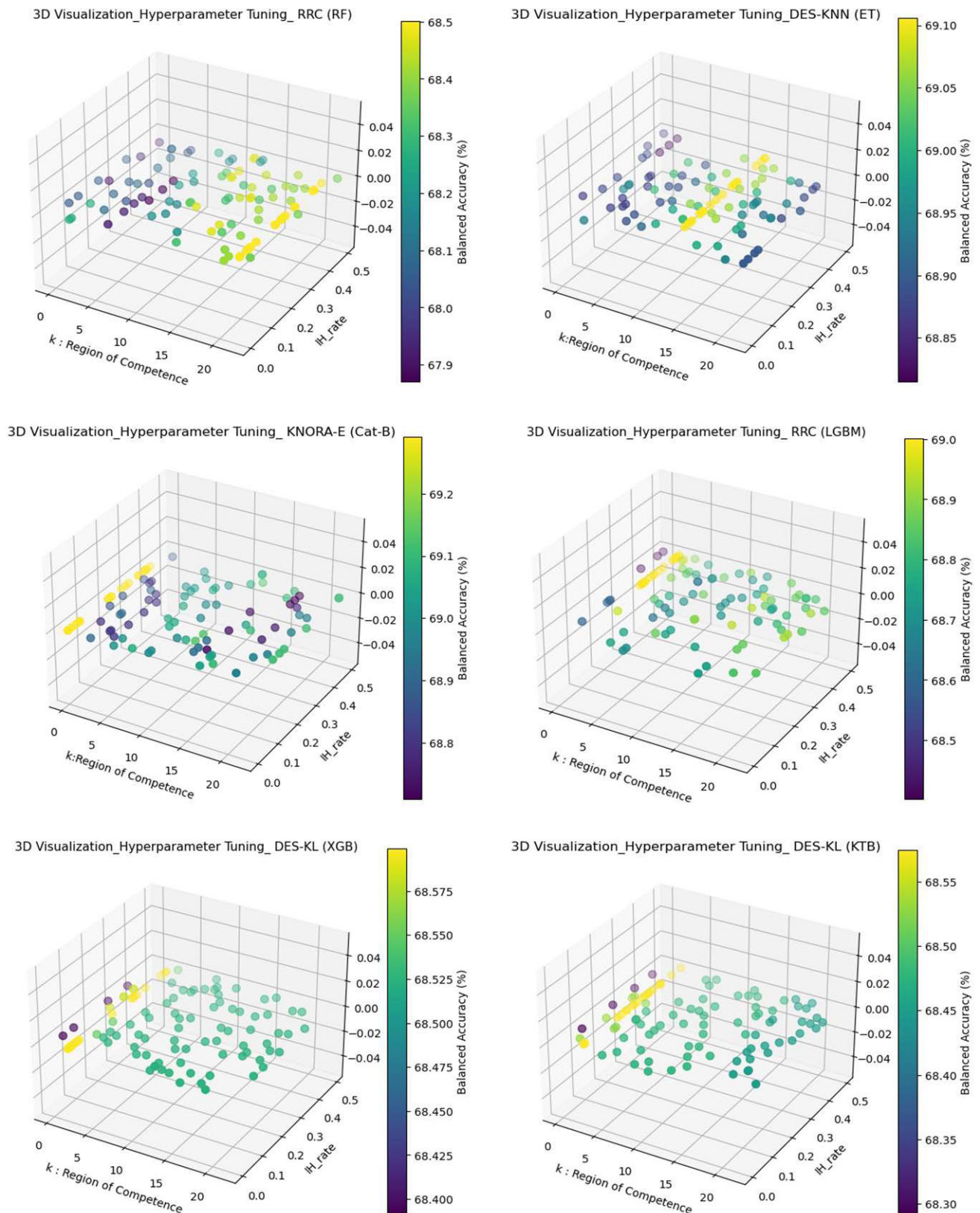


FIGURE 6. 3D scatter plot_ Enhancement of 'Main' DES methods through the integration of IH & RoC using Bayesian Optimization.

TABLE 3. Comparison of performance metrics for all static ensemble classifiers.

Base Classifiers	Balanced Accuracy	Sensitivity	MCC	G-mean
Random Forest	0.6260	0.74	0.284	0.575
Extra Tree	0.6295	0.74	0.274	0.562
Cat Boost	0.6312	0.74	0.291	0.585
Light GBM	0.6309	0.74	0.289	0.585
XG Boost	0.6265	0.73	0.279	0.580
KT Boost	0.6283	0.75	0.30	0.569

measures for all DES algorithms utilized in correspondence with the chosen base classifiers.

Upon reviewing the results, it is clear from Table 4 that the RRC consistently exhibited strong performance across various base classifiers within the DES framework. The ‘bold and underlined’ entries signify DES algorithms that outperformed others when paired with their corresponding base classifier, while ‘only bold’ entries indicate DES algorithms that ranked second in performances.

Among all the DES algorithms, as indicated in Table 4, KNORA-E ensemble with Cat Boost, DES-RRC ensemble with Light GBM, and DES-KNN ensemble with Extra Tree produced very similar outcomes, with only a slight difference in their performance matrices. However, KNORA-E with Cat Boost surpassed all DES methods in terms of performance, with a Balanced Accuracy of 69.3%, Sensitivity of 72%, MCC value of 0.36, and G-mean score of 69%. Similarly, the ensemble of DES-RRC with Light GBM secured the second position, demonstrating notable performance with a balanced classification accuracy of 69%, Sensitivity 72%, Matthew’s Correlation Coefficient (MCC) of 0.354, and a G-mean score of 68.8%. Meanwhile, DES-KNN with Extra tree as the base classifier lay third in terms of BCA at 69.1% and G-mean scores of 69%, albeit with slightly lower MCC and Sensitivity of 0.352 and 71%, respectively.

Upon examining the outcomes achieved with Random Forest, XG Boost, and KT Boost as base classifiers, it was observed that DES-RRC and KNORA-U ‘with RF as base’ exhibited commendable performance in terms of balanced classification accuracy and MCC scores. Specifically, when XG Boost served as the base classifier, DES-RRC secured the top position, followed by KNORA-U in second place. Furthermore, DES-KL demonstrated superior performance when KT Boost was employed as the base classifier, surpassing other DES methods in terms of performance when paired with KT Boost ensembles.

However, META-DES, while demonstrating promising performance in some studies, did not exhibit comparable performance measures to other DES methods across all base classifiers investigated in our study.

In general, DES-RRC demonstrated commendable performance across all base classifiers in DES, however, KNORA-E with Cat Boost as the base classifier outperformed all other DES models. Consequently, it stands as a promising choice,

TABLE 4. Comparison of performance metrics for all dynamic ensemble classifiers in conjunction with base classifiers.

DES Algorithms	Balanced Accuracy	Sensitivity	MCC	G-mean
Random Forest as Base Classifier				
KNORA-E	0.6769	0.71	0.324	0.665
KNORA-U	0.6850	0.71	0.339	0.680
DES-KNN	0.6812	0.71	0.335	0.677
DES-RRC	0.6850	0.71	0.341	0.681
DES-KL	0.6841	0.71	0.339	0.681
META-DES	0.6764	0.72	0.325	0.661
Extra Tree as Base Classifier				
KNORA-E	0.6900	0.71	0.351	0.689
KNORA-U	0.6868	0.71	0.344	0.685
DES-KNN(3rd)	0.6910	0.71	0.352	0.690
DES-RRC	0.6906	0.71	0.351	0.689
DES-KL	0.6861	0.70	0.342	0.684
META-DES	0.6842	0.71	0.340	0.680
Cat Boost as Base Classifier				
KNORA-E (1st)	0.6929	0.72	0.358	0.691
KNORA-U	0.6854	0.71	0.342	0.683
DES-KNN	0.6879	0.70	0.348	0.686
DES-RRC	0.6884	0.71	0.351	0.687
DES-KL	0.6856	0.71	0.341	0.683
META-DES	0.6864	0.72	0.347	0.682
Light GBM as Base Classifier				
KNORA-E	0.6831	0.72	0.340	0.677
KNORA-U	0.6870	0.71	0.345	0.686
DES-KNN	0.6859	0.71	0.346	0.684
DES-RRC (2nd)	0.6900	0.72	0.354	0.688
DES-KL	0.6885	0.71	0.348	0.687
META-DES	0.6817	0.72	0.344	0.676
XG Boost as Base Classifier				
KNORA-E	0.6819	0.70	0.327	0.674
KNORA-U	0.6872	0.71	0.343	0.685
DES-KNN	0.6864	0.71	0.342	0.683
DES-RRC	0.6892	0.71	0.344	0.684
DES-KL	0.6860	0.70	0.341	0.684
META-DES	0.6794	0.71	0.324	0.685
KT Boost as Base Classifier				
KNORA-E	0.6842	0.71	0.334	0.677
KNORA-U	0.6852	0.70	0.339	0.682
DES-KNN	0.6853	0.71	0.341	0.683
DES-RRC	0.6864	0.71	0.339	0.680
DES-KL	0.6857	0.71	0.341	0.683
META-DES	0.6827	0.71	0.331	0.677

particularly when coupled with SHAP analysis for feature importance assessment.

D. MODEL INTERPRETATION WITH SHAP

1) GLOBAL FEATURE IMPORTANCE AND CONTRIBUTION

One of the study’s objectives was to utilize real-time traffic crash data features to assess safety in the moment; however,

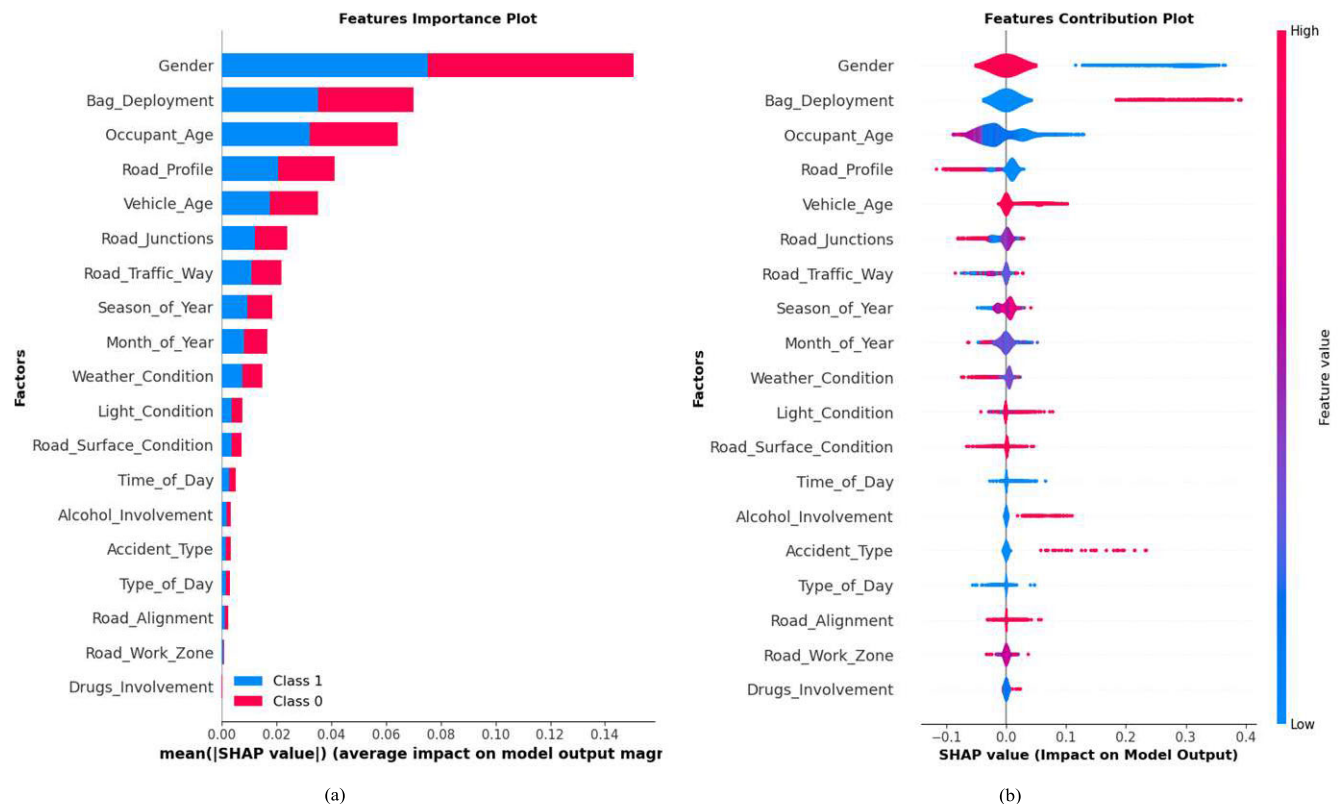


FIGURE 7. Feature importance (left) and feature contribution (right) plots with SHAP analysis.

the impact of these traffic factors on actual safety remains unclear. To conduct a thorough analysis of the effect of traffic factors on conflict likelihood, the SHAP technique is employed. The purpose of SHAP interpretation is to clarify how a machine-learning model behaves across the entire spectrum of its input factors' values. It is important to differentiate between factors' contributions and factor importance. This involves averaging SHAP values for each instance across the dataset. SHAP facilitates the assessment of the relative importance of risk factors through a SHAP factors importance plot and delineates factors' contributions via a contribution plot.

This study utilizes KNORAE-Cat Boost Model for SHAP analysis to explore the relationship between real-time crash consequences and their causes. Despite KNORA-E with Cat Boost yielding the highest prediction results, this study prioritizes the model's interpretability over its predictive ability. The effectiveness of the model's interpretation serves as the basis for feature analysis. The discussion focuses on the top 10 features because the relevance of certain features in the model is uncertain, and they may not be as significant as depicted in Fig. 7.

The significance of independent factors' importance is depicted in Fig. 7 (a). It is evident that each input parameter holds non-negligible significance, implying varying degrees of influence on expected outcomes. The analysis reveals that road user gender, bag non-deployment behavior, and user

age exert the most substantial effects on accident severity, followed by road profile, vehicle age, and road junctions. Conversely, attributes such as drug involvement, road work zones, road alignment, type of day, accident type, and alcohol involvement exhibit minimal impact on incident severity. While these findings underscore the importance of every risk factor, they do not elucidate how each contributing factor influences crash severity incidences.

The SHAP contribution image presented in Fig. 7 (b), ranks feature according to their significance in determining conflicts. Beginning with the most crucial input parameters, the features are arranged vertically based on their impact on crash severity. The color scale illustrates the level of relevance for each variable, with blue indicating weak relevance, pinkish-red indicating moderate risk, and red representing high-risk relevance. The horizontal line represents the SHAP score, while the vertical line at a SHAP value of 0.0 serves as the starting point while, a positive SHAP number indicates a higher real-time conflict risk, whereas a lower value suggests greater real-time safety. By analyzing the color distribution across the data, the effects of traffic variables can be discerned.

In the contribution plot, a majority of the light blue dots corresponding to occupant age and gender were positioned to the right of the base vertical line, suggesting that younger drivers are more susceptible to being involved in collisions resulting in serious injuries. Likewise, the red color of the

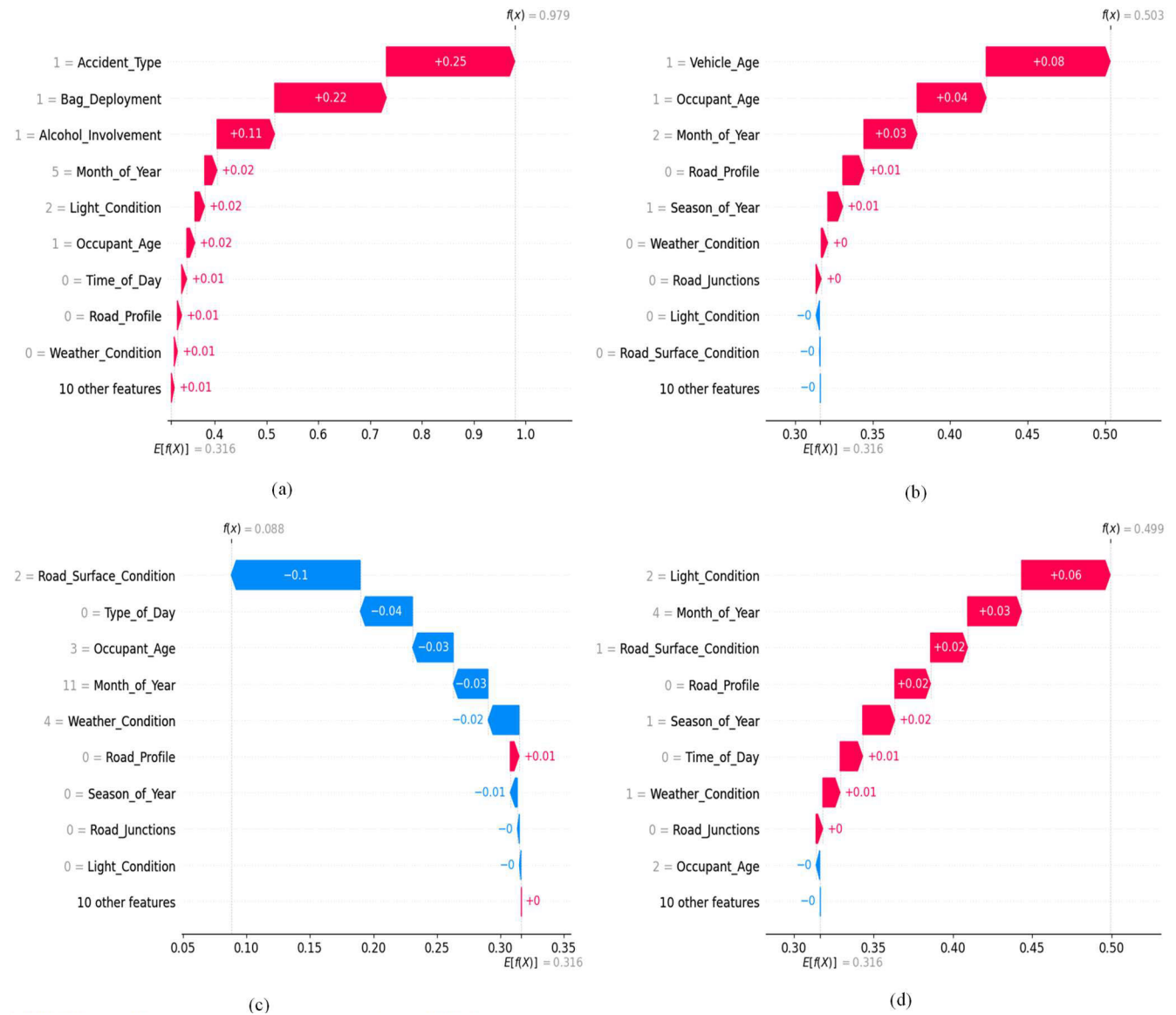


FIGURE 8. Local feature importance plot for Injury and PDO classes.

Bag deployment feature, rising positively from the baseline, indicates that the absence of airbag deployment increases the likelihood of an accident. Furthermore, a concentration of pinkish-red spots near the left side of the vertical line, with fewer blue dots on the right, for road profile, suggests that injuries are more frequent on curved roads compared to straight ones.

2) LOCAL FEATURE IMPORTANCE & CONTRIBUTION

After analyzing the importance of global features, it is essential to examine the local feature relevance for both injury and PDO cases, as shown in Fig. 8. The optimal model was utilized to predict the probability of injury and PDO for each test sample, classifying those with probabilities above 0.5 as injuries and those below 0.5 as PDO. The waterfall plot depicts an injury case with a more likely hood of injury with

a probability of 97.9% and a less likelihood of injury with a probability of 50.3%, as shown in Fig. 8 (a, b) respectively. Key features contributing to an injury having more likelihood, include alcohol involvement (1), lack of airbag deployment (1), and vehicle rollover (1), potentially resulting in serious injury outcomes.

Conversely, for an injury case Fig. 8 (b) having minor chances, prominent features include vehicle age within the range of 10-19 years, occupant age between 20 to 29 years, and accidents occurring in March. These feature values may influence the decision towards classifying the possible or minor injury. In contrast, Fig. 8 (c, d) illustrates a waterfall plot representing a case classified under the 'PDO' class.

The dominant features leading to property damage, with a 0.1% probability of injury in Fig. 8 (c), include nighttime accidents (0) on surfaces with slush or dirt, occurring in

December (11), involving an experienced road user aged between 30-39. Conversely, the property damage case in Fig. 8 (d) is characterized by prominent features such as dark lighting conditions in absence of road lights (2), slippery surface conditions leading to wet surfaces (1), and occurring in May having possibility of injury with probability of 49.9% but resulted as property damaged only. Moreover, after identifying injury cases with probabilities above the threshold of 0.5, further insights into local feature importance can be gained by examining the probabilities of injury at various levels such as 60%, 70%, and 80%.

IV. CONCLUSION AND RECOMMENDATIONS

In this study, we conducted a comprehensive comparison of six DES algorithms, namely KNORA-E, KNORA-U, DES-KNN, DES-RRC, DES-KL, and META-DES, using a pool of six input classifiers: Random Forest, Extra Tree, Cat Boost, Light GBM, XG Boost, and KT Boost. The objective was to improve crash risk assessments focusing on evaluation metrics such as Balanced Accuracy, Sensitivity, Matthews Correlation Coefficient (MCC), and G-mean score. Our investigation revealed a significant enhancement in the performance of the pool of classifiers when employing DES algorithms, particularly in scenarios involving imbalanced classification data.

The novelty of our approach lies in the utilization of DES algorithms enhanced with Instance Hardness and Region of Competence (IHRoC) considerations, introducing a novel Bayesian Optimized Dynamic Ensemble Selection strategy for crash severity forecasting. By integrating dynamic ensemble selection techniques with instance hardness and region of competence considerations, our study aimed to improve predictive performance in road traffic crash risk assessment.

Results obtained from our analysis of crash data demonstrated commendable performances across all base classifiers when utilizing Dynamic Ensemble Selection with the Randomized Reference Classifier, but particularly noteworthy was the achievement of the KNORA-E algorithm on Cat Boost, exhibiting a BCA of 69.3%, MCC of 0.36, and a G-mean score of 69.10%, on the test set. Following closely, DES-RRC with Light GBM emerged as the second-best performer. These results underscore the superiority of dynamic ensemble classifiers over static ensemble models in terms of predictive accuracy and robustness, especially when dealing with imbalanced classes.

Moreover, the study addressed the issue of transparency and interpretability inherent in machine-learning ensemble classification models. By employing SHAP analysis for the proposed KNORAE-Cat Boost model, we gained insights into the influential factors contributing to crash severity. Features such as road user gender, air-bag deployment, vehicle age, and road surface conditions were identified as significant contributors to crash severity outcomes, enhancing the transparency and interpretability of the developed

models. In summary our study contributes to the development of targeted interventions and safety measures aimed at mitigating the impact of road accidents by identifying specific features associated with different levels of crash injury severity.

As part of our future endeavors, we intend to explore multi-class imbalanced dataset, encompassing minor, major, and fatal injuries. Our focus will be on conducting an in-depth analysis of DES models, with particular emphasis on DES-RRC and DES-MI to compare their performance with ensemble imbalance learning techniques, including easy ensemble, self-paced ensemble and local cascade ensemble method.

Transportation authorities and safety researchers are advised to integrate DES techniques into their crash severity forecasting frameworks, particularly those enhanced with instance hardness and region of competence considerations. These advanced methodologies offer enhanced predictive accuracy and reliability compared to conventional static ensemble methods. Furthermore, there is a pressing need for continued exploration of model interpretability techniques to unravel the complex interplay between various risk factors and crash severity outcomes.

Future research endeavors should prioritize the incorporation of local feature importance analysis into crash severity prediction models. This analytical approach yields actionable insights that can inform policy decisions, infrastructure planning, and emergency response strategies aimed at improving road safety. Collaborative efforts between researchers, transportation agencies, and industry stakeholders are essential for the validation and deployment of validated models for real-time crash severity forecasting and decision-making. Moreover, public awareness campaigns targeting drivers, pedestrians, and other road users should be intensified to emphasize the importance of road safety and individual responsibility in mitigating crash severity. These initiatives play a crucial role in fostering a culture of compliance with safety regulations and reducing the frequency and severity of road accidents.

Before practical implementation, it is imperative to subject the developed methodology to rigorous validation and testing using real-world crash data from diverse geographical locations and traffic environments. Additionally, continuous refinement of predictive models based on updated data and evolving risk factors is essential to ensure the accuracy and reliability of crash severity forecasts.

DATA AVAILABILITY STATEMENT

Supporting data for the study's conclusions may be obtained from the corresponding author upon reasonable request.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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